T2D Digital Twin

Forecasting high and low glucose events 2 hours ahead by fusing EHR and wearable data

2 h

early warning window

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Diabetes care in India is still reactive

101 M

people living with diabetes in India (ICMR-INDIAB, 2023)

Blind spots

glucose spikes between clinic visits go unseen

2 h

ahead: the warning window this twin aims for

Goal

A virtual replica of the patient that predicts adverse glycaemic events before they happen, and lets the doctor test lifestyle changes virtually.

A digital twin fuses two data streams

Static: EHR

- Age, sex, BMI
- HbA1c, fasting glucose, eGFR
- Diabetes duration, medication
- Genetic marker (TCF7L2)

Twin model

2-hour risk and what-if

Dynamic: wearable

- Glucose (CGM), every 15 min
- Heart rate
- Steps
- Nightly sleep hours

Focus condition: Type 2 Diabetes

Privacy-safe synthetic data

300

virtual patients

14 days

of monitoring each

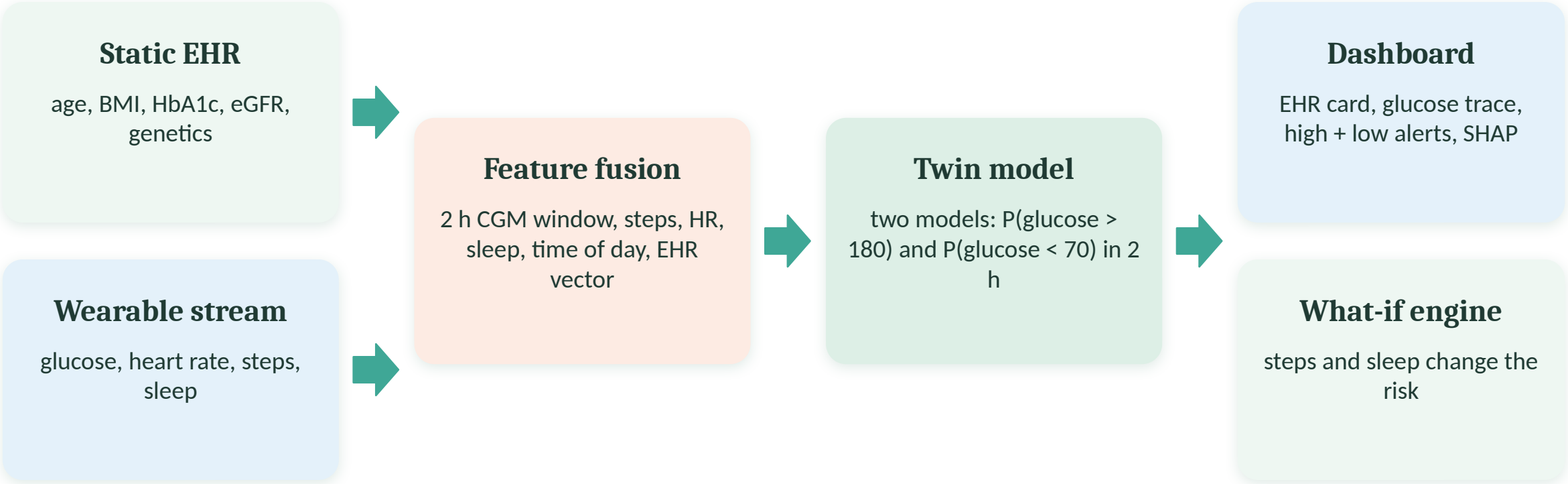
15 min

sampling resolution

How the simulator works

- Meal-driven glucose spikes from Indian-diet carbohydrate loads
- Scaled by insulin resistance from BMI, HbA1c, TCF7L2 alleles and metformin
- Walking lowers spikes; short sleep raises them; drug dose + skipped meals cause lows
- No real patient data used, so no personal-data exposure

System architecture



Evaluation: ablation of CGM only vs + wearable vs + EHR, with ROC-AUC, PR-AUC and SHAP

How the twin learns

8

CGM window features: level, spread, trend over 2 h

7

wearable and time features: steps, heart rate, sleep, hour

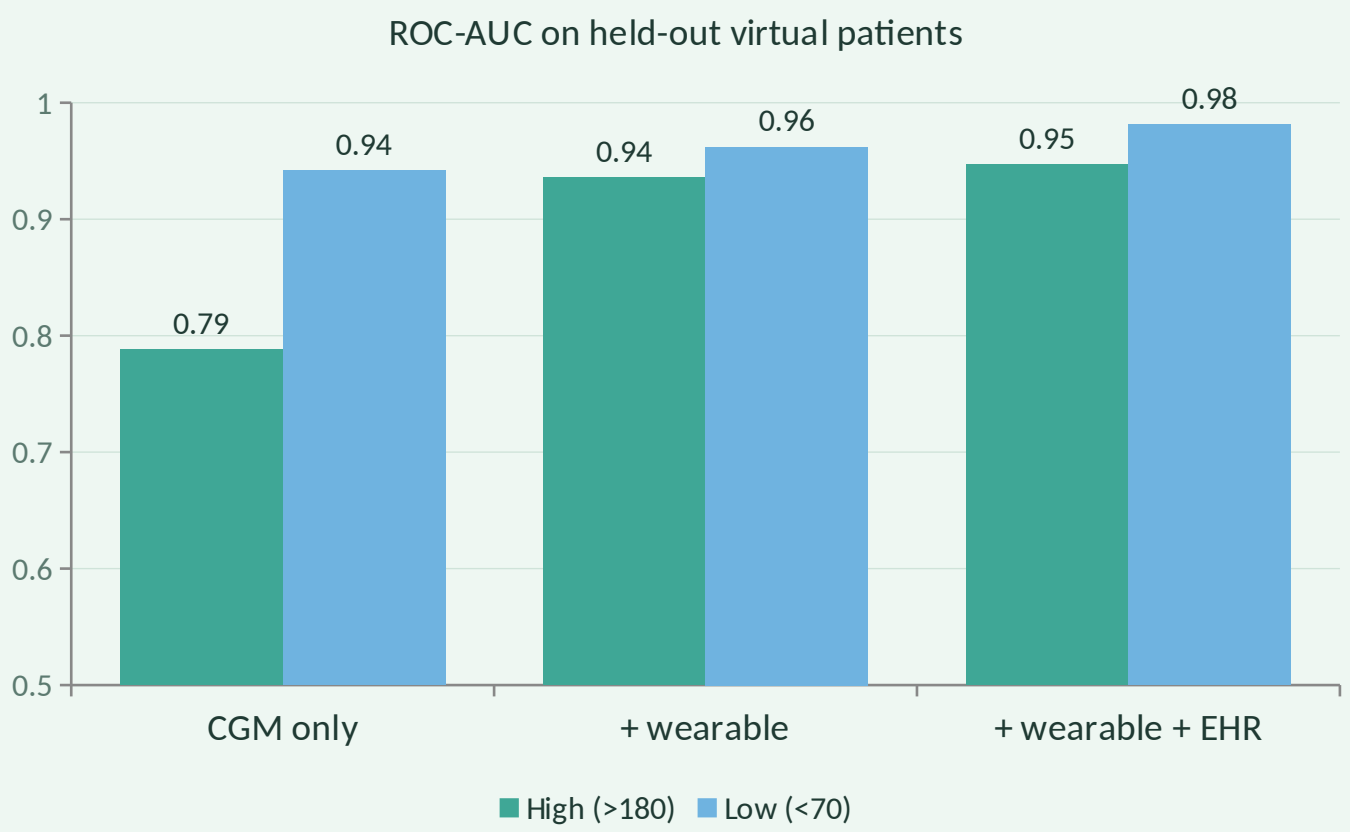
12

EHR features: demographics, labs, genetics, medication

Target and training

Two gradient boosting models: will glucose go above 180 mg/dL, or below 70 mg/dL, within the next 2 hours? Trained on 75% of patients, tested on unseen 25% (no leakage).

Fusing both streams wins, for highs and lows



0.947

ROC-AUC, high-glucose twin

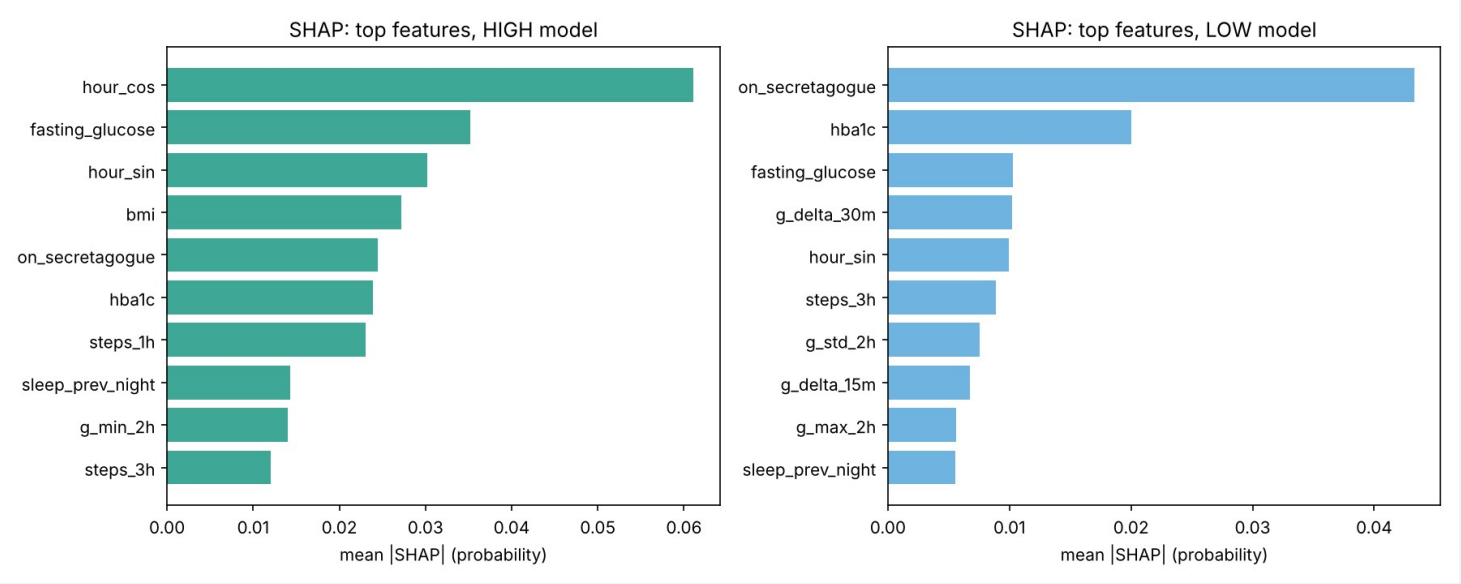
0.981

ROC-AUC, low-glucose twin

Lows are rare (2%): PR-AUC 0.61 vs 0.02 by chance. Synthetic data: an upper bound, not clinical evidence

Why did it alert? SHAP explanations

Global view: which features matter most



Left: high model. Right: low model.

What SHAP does

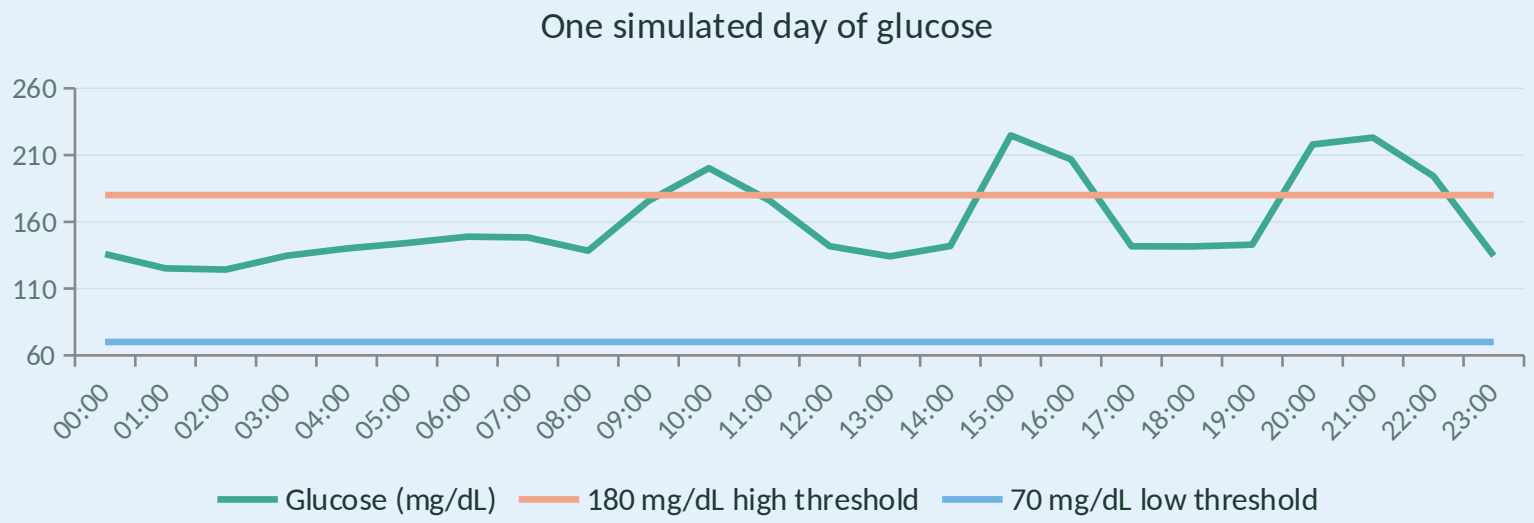
Shares credit for one prediction among the features, like splitting a team prize fairly. The dashboard shows the push of each feature for the chosen patient and moment. It explains the model, not biology.

Doctor's view of the virtual patient

Patient #3

EHR card: age, BMI, HbA1c, eGFR, medication, genetic risk, with live glucose, sleep and steps

Separate HIGH and LOW alerts with adjustable thresholds



What-if simulation

Slide steps or sleep up or down and the twin re-scores both 2 hour risks instantly.

Clinical safety thinking (SaMD)

Intended use

Clinician decision support for adults with T2D wearing a CGM. Not for treatment, dosing or replacing device alarms.

Likely SaMD class

Software as a Medical Device, plausibly low-to-middle risk (IMDRF I to II). Needs a formal regulatory assessment (CDSCO in India).

Top risks

Missed low alert, false alarms, synthetic-data bias, what-if misread as advice. Each has a mitigation in SAFETY.md.

No compliance claim

No ISO 13485 / 14971 / IEC 62304 evidence exists here, so none is claimed. Real validation comes first.

Why it matters, and what to be honest about

Relevance to India

- Huge and growing T2D burden
- Carbohydrate-heavy diets
- Low-cost wearables and CGM are spreading
- Proactive alerts support earlier intervention

Limitations

- Synthetic data: results are an upper bound
- Simulator encodes assumed physiology
- Lows depend on a simulated drug effect
- Not a medical device; not clinically validated

Next steps

1

Validate on real data

ShanghaiT2DM test script included; then Indian cohorts

2

Richer signals

HRV, sleep stages, meal logging, Indian food database

3

Better models

LSTM / TCN sequence models, SHAP explanations

4

Clinician pilot

Test usefulness and alert fatigue with doctors

Thank you